Paper Code : 21306

F-406

B.C.A. (Second Semester)

Examination, 2022

(New Course)

Paper No. BCA-N-201

DIGITAL ELECTRONICS

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Fill in the blanks : 4

(i) $(1010.011)_2 = (\ldots\ldots\ldots)_{10}$

(ii) $(0.513)_{10} = (\ldots\ldots\ldots)_8$

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(iii) $(B65F)_{16} = (\ldots\ldots\ldots)_{10}$

(iv) $(153)_{10} = (\ldots\ldots\ldots)_8$

(b) What are universal gates? Why are they called universal? 3

(c) Explain with the help of block diagram the working principle of a full adder. 4

(d) Draw 2-input OR Gate using one 4×1 MUX. 3

2. (a) What is a flip-flop? What are the uses of flip flops? Explain it? 7

(b) Define computer organization, computer design and computer architecture. 7

3. (a) Explain the working of catch memory. 3

(b) Explain the concept of virtual memory. 4

(c) Explain the various addressing modes used in computer system. 7

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4. (a) Design a 4-bit up-down counter. 7

(b) What is Register? Explain serial Register in detail. 7

5. (a) Differentiate b/w Combinational Digital circuits and sequential digital circuit. 7

(b) Distinguish b/w the following : 7

(i) RAM and ROM

(ii) Static and Dynamic memory

6. Write short notes on the following : 14

(a) CPU

(b) Demultiplex

(c) Adder

(d) Serial communication

7. (a) Show the logic diagram of clocked RS flip-flop with four NAND Gates. 7

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(b) Show the logic diagram of a clocked D flip-flop with AND and NOR Gates. 7

8. (a) What is the difference between serial and parallel transfer? What type of register is used in each case? 7

(b) Discuss various bus control methods. 7

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